Reg.No:……
Name:…….

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BBA (HONOURS)
FOURTH SEMESTER EXAMINATION (2024 ADMISION ONWARDS)
MG4SECBBA200 –STATISTICS FOR BUSINESS
MODEL QUESTION PAPER

Duration: 1.5 hrs **Maximum Marks: 50**

Part A
Short Answer Type Questions
Answer any 5 out of 6 Questions. Each question carries 3 marks.

1. Give three limitations of statistics. [U][CO1]
2. Write the additive model of a Time series. [K][CO2]
3. What is a time series? [U]CO2]
4. Define seasonal variation. [U][CO2]
5. What is a contingency table? [U][CO3]
6. Distinguish between Type 1 and Type II errors. [K][CO3]

[5x3 = 15 marks]

Part B
Short Essay Type Questions
Answer any 3 out of 4 Questions. Each question carries 5 marks.

7. What are the components of time series? Explain.[An][CO2]
8. Fit a trend line for the following data by semi average method.[An] [CO2]

Year	2018	2019	2020	2021	2022	2023	2024	2025
Sales(in lakhs)	12	14	16	22	20	31	28	17

9. Ten soldiers visit a rifle range for two consecutive weeks, for the first week, their scores are 67,24,57,55,63,54,56,68,33,43 and during the second week, they score in the same order 70,38,58,58,56,67,68,72,42,38. Examine if there is a significant difference in their performance. [A][CO3]
10. Explain Friedman's two way analysis of variance in testing of hypothesis.[U][CO3]

[3x5=15 marks]

Part C
Essay Type Questions
Answer any 2 out of 4 Questions. Each question carries 10marks.

11. Explain the scope of statistics in
    a) Business management b) Business and Industry c) Research d) Economics [U] [CO1]

12. From the following data use chi-square test and conclude whether innoculation is effective in preventing tuberculosis[A][CO3]

	Attacked	Not attacked
Innoculated	20	300
Not innoculated	80	360

13. Fit a straight line trend to the following data by the method of least squares. Estimate the most likely estimated production for the year 2026. [An] [CO2]

Year	2019	2020	2021	2022	2023	2024	2025
Production (‘000 tonnes)	10	13	12	14	12	16	14

14. A research is concerned about the IQ level of students from different departments in a school. Data of an IQ test result (in percentage) is recorded. Is there any significant difference between the IQ levels of students in different departments? [A][CO3]

Science	44	56	78	85
Arts	58	25	65	73
Commerce	65	45	58	38

[2x10 =20 marks]